

Sustainability and high-level corruption in healthcare procurement: Profiles of Italian contracting authorities

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ABSTRACT

Addressing corruption is crucial for building a sustainable healthcare system that ensures access, quality, and equity in healthcare delivery. Despite that, many current strategies to combat corruption in the healthcare sector do not evaluate high-level corruption, such as corruption risks occurring at sub-national levels. This work bridges this gap by providing corruption risk profiles of Italian contracting authorities responsible for procuring goods and services for healthcare facilities in the public procurement process. Using an array of 14 red flags of corruption risk and an extended Item Response Theory model applied to a big data source made available by the Italian Anti-corruption Authority, our main findings show that: *i.* the risk of corruption is a multidimensional occurrence, which can be represented as a four-dimensional latent variable; *ii.* there are eight clusters of contracting authorities, having distinct and well-defined risk profiles over the four ascertained dimensions of corruption risk; *iii.* the distribution of risk profiles at sub-national level showcases relevant geographic variations and emphasises the need for tailored anti-corruption strategies to effectively address region-specific challenges and risk factors.

1. Introduction

Corruption in the health sector can undermine the principles of sustainability by hindering resource allocation, access to healthcare, and the quality of care. When funds are misappropriated or embezzled, corruption can lead to inadequate healthcare services, thereby affecting the overall health and well-being of the population. Moreover, corruption can erect barriers to accessing healthcare services. In cases where bribes or kickbacks are necessary to receive medical care, unequal access to healthcare can arise. Further, corruption may lead to the proliferation of substandard or counterfeit medical products and services, jeopardising patient safety and, in the long run, negatively impact the sustainability of a healthcare system striving to provide quality care.

Despite recent advancements in developing tailored strategies to combat corruption in the healthcare sector, numerous existing anti-corruption approaches overlook high-level corruption, particularly risks at sub-national levels. This paper aims to address this gap by presenting corruption risk profiles of Italian contracting authorities involved in procuring goods and services for healthcare facilities and by illustrating their regional distribution.

Several factors contribute to the vulnerability of health systems to corruption [1]. The substantial financial resources flowing through

health systems worldwide create opportunities for abuse and illicit gains. Health spending varies widely, ranging from around 4% of GDP in low-income countries to over 15% in OECD countries. Health systems also suffer from a high degree of information asymmetry, where information is not equally accessible to all stakeholders. Patients often lack the information needed to assess medical decisions, bills, and the necessity of services. This information imbalance also affects procurement decisions, as pharmaceutical companies often possess more information than doctors or regulators. The involvement of numerous dispersed actors in health systems (including medical suppliers, providers, payers, and policymakers) exacerbates difficulties in transparency and conflict of interest detection. Moreover, national health systems often exhibit complexity and fragmentation, characterised by diverse subsystems catering to different populations and involving a mix of public and private providers and insurers. Beyond the complexity of health systems, the globalised supply chain for drugs and medical devices, driven by business goals, may conflict with the public's medical needs.

The development of tailored approaches to address corruption in the healthcare sector is relatively recent. The initial set of policies and program guidelines targeting the health sector began to emerge around 2010. This included the widely known U4 Issue, titled "Addressing corruption in the health sector: Ensuring fair access to healthcare for

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all” [2]. Subsequently, anti-corruption efforts with a healthcare focus have been undertaken on a global scale. Among them, some assessment tools are specifically designed to address perceptions or direct experiences of corruption – such as the Transparency International Corruption Perceptions Index, the World Bank Worldwide Governance Indicator, the Transparency International Global Corruption Barometer – while others take a broader view of healthcare sector governance [3], or focus on specific aspects or sub-sectors within healthcare, such as pharmaceuticals [4].

Most recently, the adoption of the 2030 Agenda for Sustainable Development marks a significant advancement in the fight against corruption [5]. For the first time, it establishes a clear connection between reducing corruption and fostering peaceful, equitable, and inclusive societies by emphasising the importance of promoting transparency, accountability, and anti-corruption measures. Nonetheless, the 2030 Agenda does not explicitly address instances of corruption in the healthcare sector. While addressing corruption within the broader context of governance and institutions can undoubtedly benefit various public sectors (including healthcare), Sustainable Development Goal 3 (SDG 3), which focuses on “Good Health and Well-being”, does not include specific indicators aimed at combating corruption in the health sector. Similarly, SDG 16, dedicated to “Peace, Justice, and Strong Institutions”, also lacks health-specific indicators.

This work aims to close the above divide by measuring high-level corruption in the Italian public procurement healthcare sector over the last ten years (2013–2022). By relying on an array of 14 well-established red flags of corruption risk in public procurement, we profile contracting authorities responsible for procuring goods and services for healthcare facilities in the public procurement process. Contracting authorities are categorised into sub-groups, each characterised by various dimensions of interrelated corruption risks. To these purposes, we use a big data source made available by the Italian Anti-corruption Authority (ANAC) and rely on an extended Item Response Theory (IRT) model, the latent class multidimensional IRT model [6,7]. This model concurrently accounts for *i.* the latent nature of corruption risk, as it cannot be directly observed in the data due to its hidden nature; *ii.* the multidimensional facet of corruption risk and the complex relational structure of our data (i.e., interrelated red flags); *iii.* the identification and qualification of profiles of contracting authorities as regards the ascertained corruption risk dimensions.

The paper is organised as follows. Section 2 briefly describes available methods for measuring corruption, while data, indicators and statistical approach employed for this work are presented in Sections 3 and 4. Results are then shown in Section 5, whose discussion is provided in Section 6.

2. Measuring approaches to combat corruption

Corruption, akin to several other factors considered in the pursuit of sustainable development, manifests as multifaceted and latent phenomenon, presenting significant challenges in both its measurement for repression and prevention purposes. Involving various interconnected economic and social aspects, corruption may yield indirect and long-term consequences that are not immediately observable or measurable, owing to its latent nature.

Current tools used to measure corruption are diverse and encompass a wide range of approaches. These include measures based not only on reported direct experiences of corruption but also on subjective perceptions, judicial-based measures, statistical inference proxies, and corruption risk indicators [8]. In the 2030 Agenda, SDG 16 contains two key targets particularly relevant for the context at issue: *i.* Target 16.5, devoted to substantially reduce corruption and bribery in all their forms, and *ii.* Target 16.6, devoted to promoting effective, accountable, and transparent institutions. The monitoring of Target 16.5 is assigned to two indicators: direct experiences of corruption and bribery by individuals (Indicator 16.5.1) and businesses (Indicator

16.5.2), in their interactions with public officials. Primary government expenditures, computed as a proportion of original approved budget (Indicator 16.6.1), and the proportion of population satisfied with their last experience of public services (Indicator 16.6.2) are chosen within Target 16.6.

Direct measures of corruption, such as Indicator 16.5.1 and Indicator 16.5.2, offer valuable insights into actual experiences of corruption but face challenges related to under-reporting. Victims or businesses may hesitate to report corruption due to fear of reprisal or lack of trust in the reporting mechanisms. On the other hand, perception-based measures, widely used in comparative studies, may not accurately reflect factual involvement in corruption crimes [9]. What is more, these measures can be influenced by factors not directly related to corruption, such as the general sentiments reflecting prior economic growth [10].

In recent years, various proposals have surfaced to mitigate the shortcomings of both direct and subjective measures of corruption. These include experimental methodologies and indirect proxy measures, such as corruption risk indicators, commonly known as red flags of corruption risks. They gauge vulnerabilities or anomalies that may signal potential corrupt practices within an organisation, project, or system. The identification and monitoring of red flags can assist organisations and control authorities in proactively addressing corruption risks and implementing appropriate preventive measures to tackle it [8,11–13].

One of the government areas where corruption is most prevalent is public procurement, making red flags a common tool for assessing corruption risks within the public procurement cycle. In the public procurement process – through which public authorities acquire goods, products, or services from businesses – several occurrences can signal potentially risky situations [14]. These include delays or failures in contract award, non-open or exceptional procedure forms such as direct awards, contract extensions and/or modifications over time, and single bidding, i.e., the circumstance that a procedure receives a single offer by a single bidder [15,16].

Previous studies [17] demonstrate that corruption risk is not a one-dimensional phenomenon, but rather a multidimensional latent occurrence characterised by diverse behaviours, facts, and circumstances that vary in nature, type, and magnitude. In practical terms, this suggests that single measures of corruption risk, such as red flags, cannot be expected to relate with each other. Instead, they may correlate within sub-groups, each representing a distinct and specific risk category within the broader spectrum of corruption risk. These sub-dimensions of corruption risk, each comprising a subset of highly correlated red flags, may relate to excess of discretionary power, lack of competition, or inefficiencies in the public procurement process. Each of these factors can have detrimental effects on the health system, leading to sub-optimal outcomes in terms of cost-effectiveness and service quality. Additionally, they may result in missed opportunities for cost savings and innovation, as well as erosion of trust in the health system.

3. Dataset and red flags of corruption risk

This study relies on a set of 14 red flag indicators computed at the contracting body level [17], as listed in Table 1. They are computed by aggregating information from public contracts overseen by each contracting authority within the period 2013–2022. The information about all the public procedures managed by Italian contracting authorities are included in the Italian National Database of Public Contracts (abbreviated as BDNCP, standing for *Banca Dati Nazionale dei Contratti Pubblici*). This dataset is managed by the Italian anticorruption authority (ANAC) and is released in an open version.¹ It is organised into multiple tables,

¹ <https://dati.anticorruzione.it/opendata>

Table 1

List of selected red flags, computed at the contracting authority level.

	Label	Metric	Description
1	<i>nonopen_num</i>	Proportion	Number of non-open procedures (direct awards and negotiated procedures) out of all the procedures
2	<i>nonopen_econ</i>	Proportion	Economic value of non-open procedures out of the economic value of all the procedures
3	<i>singlebid_num</i>	Proportion	Number of procedures with a single bid out of all the procedures
4	<i>singlebid_econ</i>	Proportion	Economic value of procedures with a single bid out of the economic value of all the procedures
5	<i>MEAT_num</i>	Proportion	Number of procedures awarded using MEAT criterion out of all the procedures
6	<i>MEAT_econ</i>	Proportion	Economic value of procedures awarded using MEAT criterion out of the economic value of all the procedures
7	<i>advert</i>	Days (on average)	Average advertising period (from the call for tenders publication to the bid submission deadline)
8	<i>eval</i>	Days (on average)	Average bid evaluation period (from the bid submission deadline to the award notice)
9	<i>excl_bids</i>	Proportion	Average proportion of excluded bids over all the procedures
10	<i>all_bids_excl_but_one</i>	Proportion	Number of procedures with all bids excluded but one out of all the procedures
11	<i>excl_bids_but_one</i>	Proportion	Average proportion of excluded bids over all the procedures with all bids excluded but one
12	<i>variants</i>	Proportion	Number of procedures with at least one variant out of all the procedures
13	<i>amount_dev</i>	Ratio (on average)	Deviation in contract cost (ratio between award and cleared value) averaged over all the procedures
14	<i>time_dev</i>	Ratio (on average)	Deviation in contract duration (ratio between real and envisaged duration) averaged over all the procedures

MEAT: most economically advantageous tender.

Table 2

Distribution of selected public contracts by year and main object (year-specific percentage composition in parentheses).

Publication year	Goods	Works	Services	Total
2013	83,091 (85.9)	1,428 (1.5)	10,158 (10.5)	96,690
2014	83,459 (87.3)	1,363 (1.4)	8,710 (9.1)	95,546
2015	94,584 (88.2)	1,435 (1.3)	9,170 (8.6)	107,204
2016	85,933 (88.0)	1,102 (1.1)	8,582 (8.8)	97,633
2017	114,146 (89.1)	1,368 (1.1)	10,556 (8.2)	128,087
2018	123,307 (89.8)	1,356 (1.0)	10,583 (7.7)	137,264
2019	100,498 (87.1)	1,520 (1.3)	11,361 (9.8)	115,398
2020	118,148 (88.0)	2,148 (1.6)	11,968 (8.9)	134,284
2021	105,490 (86.6)	2,229 (1.8)	12,065 (9.9)	121,805
2022	117,058 (87.3)	2,167 (1.6)	12,914 (9.6)	134,161
Total	1,025,714 (89.4)	16,116 (1.4)	106,067 (9.2)	1,147,897

each corresponding to a distinct stage of the procurement process, such as call for tenders, award notice, contract start and end, amendments, etc. [18]. Each stage-specific table contains detailed information about tenders that have progressed to the selected phase. Tables can be merged using two keys: CIG and award ID. The CIG, which stands for *Codice Identificativo Gara*, is the contract unique identifier and is assigned to each procedure during the call for tenders phase. The award ID is acquired when the contract progresses to the award phase, allowing access to information related to the stages following the award notice.

To narrow down the healthcare authorities' market, we select public procedures from the BDNCP, by focusing on contracts managed by specific contracting authorities, that is, local health authorities, hospital administrations, and university hospital administrations, using their ID code (*codice fiscale*). In total, we select 207 contracting authorities, which are involved in approximately 1.1 million public contracts.

An overall summary of these contracts is presented in Table 2. Nearly 90% of the contracts managed by healthcare contracting authorities pertain to the provision of goods, with approximately 9% involving service purchases. This composition remains fairly constant across years. Furthermore, a consistent share of contracts used for good supply is represented by Common Procurement Vocabulary (CPV) codes starting with 33 (91.5%), which includes "Medical equipments, pharmaceuticals and personal care products". This is followed by CPV

codes starting with 30 (1.8%) and 39 (1.0%), which pertain to "Office and computing machinery, equipment and supplies except furniture and software packages" and "Furniture, furnishings, domestic appliances and cleaning products", respectively. Regarding contracts concerning service provision, the most represented categories are those with CPV codes starting with 85 ("Health and social work services"), 50 ("Repair and maintenance services"), and 72 ("IT services: consulting, software development, Internet and support"). These categories account for 21.7%, 20.7%, and 14.2% of all contracts for services, respectively.

The 14 red flags are then computed using the above data, in which two types of missingness issues may arise. Firstly, if a single contract lacks the necessary information for a specific indicator computation, that contract is excluded from the processing of that indicator (which, we recall, are considered at the contracting authority level). For example, consider the single bidding indicator (indicator 3 or 4, see Table 1): if one or more procedures for a given contracting authority have not the information about the number of submitted/excluded bids, those contracts are not considered when computing the corresponding red flag(s).

The second issue related to missingness arises when there are missing values in the red flags computed at the contracting body level, due to absence of available contracts to compute that red flag. These cases are quite rare in our data due to the extended time span (ten years) and the specific type of administrations (healthcare authorities) considered. In this regard, we can refer to Table 3, which shows descriptive statistics about the selected red flags computed at the contracting authority level. As we can see, only three red flags have missing values, albeit to a small extent.

All the indicators consist of proportions (hence, ranging between 0 and 1), with the exception of red flags 7 and 8, which measure average time extensions (in days), and red flags 13 and 14, which consist of average ratio between two quantities (contracts amounts or durations). In the following, as the red flags are presented either as proportions or means, we convert them into ordinal indicators on a uniform scale with an identical number of categories. Each category signifies a distinct level of corruption risk. Specifically, red flag indicators are discretised into four categories reflecting escalating corruption risks: *i.* no or low risk, *ii.* medium-low risk, *iii.* medium-high risk, and *iv.* high risk. Thresholds are determined by relying on the quantiles of each observed indicator distribution, as there is no universally accepted and objective guideline for defining corruption risk categories [17].

These categorical indicators are then used as the basis for subsequent analyses with the aim of concurrently *i.* evaluating the dimensionality structure of these red flags (and of the corruption risks of which they are proxies) and *ii.* extracting profiles of healthcare contracting authorities on account of the ascertained dimensions of corruption risk.

Table 3
Descriptive statistics of the red flags computed at the health authority level.

Red flag	Min	Q1	Median	Mean	Q3	Max	Prop. missing
<i>nonopen_num</i>	0.092	0.514	0.710	0.676	0.846	1.000	–
<i>nonopen_econ</i>	0.027	0.271	0.396	0.416	0.551	1.000	–
<i>singlebid_num</i>	0.258	0.559	0.657	0.648	0.750	0.953	–
<i>singlebid_econ</i>	0.032	0.411	0.545	0.534	0.643	0.928	–
<i>MEAT_num</i>	0.003	0.075	0.140	0.176	0.237	0.783	–
<i>MEAT_econ</i>	0.002	0.116	0.216	0.234	0.331	0.767	–
<i>advert</i>	9.5	21.7	28.2	28.8	35.4	68.3	–
<i>eval</i>	19.0	69.9	91.3	117.2	154.9	439.6	–
<i>excl_bids</i>	0.000	0.024	0.039	0.045	0.059	0.177	–
<i>all_bids_excl_but_one</i>	0.000	0.026	0.050	0.064	0.086	0.313	–
<i>excl_bids_but_one</i>	0.000	0.567	0.595	0.575	0.625	0.798	–
<i>variants</i>	0.000	0.012	0.041	0.083	0.130	0.458	0.019
<i>amount_dev</i>	0.576	0.936	1.000	0.980	1.042	1.281	0.232
<i>time_dev</i>	0.714	1.014	1.072	1.115	1.161	2.094	0.068

4. Extended item response theory models

The utilisation of Item Response Theory (IRT) models represents an invaluable methodological approach, particularly in cases where the data exhibit a latent structure, such as the context of our study, where corruption serves as a non-directly observable variable. In particular, we employ a variant of traditional IRT models, which incorporates two significant distinctions. Firstly, we assume that the latent structure underlying the data is multidimensional, acknowledging the existence of multiple latent factors influencing the observed responses. Secondly, our model hinges on latent variables that follow a discrete distribution, implying that the population under investigation can be divided into a finite number of distinct groups or classes. Within each of these classes, statistical units share a common level of the latent variable(s), leading to a categorisation of units with similar characteristics. Specifically, we rely on a class of multidimensional IRT models that deal with ordinal responses [7], thereby expanding on the work of Bartolucci for dichotomously-scored indicators [6]. Consequently, we refer to this class of model as the multidimensional latent class IRT model for ordinal responses.

The term “multidimensional”, as outlined above, implies that the available indicators contribute to measure, for example, D dimensions or latent traits, which can be included in latent vector denoted by $\theta = [\theta_1, \theta_2, \dots, \theta_D]^T$, with a possible realisation denoted by $\theta = [\theta_1, \theta_2, \dots, \theta_D]^T$. Moreover, the term “latent class” implies that θ is assumed to have a discrete distribution characterised by k support points, denoted by $\{\theta_1, \theta_2, \dots, \theta_k\}$, and associated mass probabilities $\{\pi_1, \pi_2, \dots, \pi_k\}$, that is, $\pi_c = P(\theta = \theta_c)$. This assumption is akin to hypothesising that the underlying population can be segmented into a limited number of hidden groups (k in this case), or latent classes. In fact, units belonging to latent class c share identical levels of latent traits, represented by the elements of vector $\theta_c = [\theta_{c1}, \dots, \theta_{cD}]^T$, with $c = 1, \dots, k$.

The proposed modelling approach establishes a probabilistic relationship between the observed values taken by red flags (i.e., manifest variables) and specific features of statistical units involved (clustered in latent classes) and red flags themselves. In particular, let X_j denote the variable associated with red flag j , with $j = 1, \dots, J$, where J indicates the total number of indicators (or items). Each red flag is assumed to have the same number of ordinal categories, denoted by h , although it is possible to have a different number of categories for each indicator.

The conditional probability of observing category x in indicator j for units with generic latent trait level θ , $P(X_j = x|\theta)$, is shortly denoted by $\phi_{j,x}(\theta)$, with $j = 1, \dots, J$ and $x = 0, \dots, h-1$. The IRT framework proposes the following equation for modelling this conditional probability:

$$f_x[\phi_j(\theta)] = \gamma_j \left(\sum_{d=1}^D \delta_{jd} \theta_d - \beta_{jx} \right), \quad (1)$$

where $f_x(\cdot)$ is a category-specific link function, $\phi_j(\theta)$ is the vector of conditional probabilities of indicator j including the single probabilities

for each category, from $x = 0$ to $x = h-1$ (hence, summing up to 1), and γ_j and β_{jx} are indicator-specific parameters, measuring its discrimination and difficulty, respectively. Moreover, the equation introduces a binary variable, denoted as δ_{jd} , which is set to 1 when item j assesses latent trait of type d , and 0 in other cases, with $j = 1, \dots, J$ and $d = 1, \dots, D$.

As regards missing values in some indicators, unlike the classical test theory – which excludes entire observations with at least one missing item – IRT models handle missing data by not scoring missing items, thus mitigating the risk of introducing bias. This approach leverages the fact that missing responses do not contribute to the item measure and is consistent with the concept of ignorable missingness. Under ignorability, IRT parameter estimates obtained using maximum likelihood methods are solely reliant on the observed responses [19,20].

Various IRT models for ordinal indicators arise depending on the type of link function (e.g., based on local or global logits). In this study, following an assessment of model fitting with different link functions using the available data, we opt for a link function based on global logits (also known as cumulative logits). Thus:

$$f_x[\phi_j(\theta)] = \log \frac{P(X_j \geq x|\theta)}{P(X_j < x|\theta)} = \frac{\phi_{j,x}(\theta) + \phi_{j,x+1}(\theta) + \dots + \phi_{j,h-1}(\theta)}{\phi_{j,0}(\theta) + \phi_{j,1}(\theta) + \dots + \phi_{j,x-1}(\theta)}.$$

This allows us to use a multidimensional extension of the (latent class) graded response model [21].

To ensure the model identifiability, it is necessary to impose appropriate constraints on the parameters. Thus, in Eq. (1) and for each dimension d , we constrain one discrimination index to 1 and one difficulty parameter to 0. Consequently, the proposed model involves a set of free parameters (m) that considers *i.* prior probabilities π_c , *ii.* the number of support points θ_{cd} , and *iii.* the count of unconstrained discrimination and difficulty parameters, with $c = 1, \dots, k$ and $d = 1, \dots, D$:

$$m = (k-1) + kD + (J-D) + (h-1)J - D.$$

The proposed model can be effectively estimated using a procedure based on the EM algorithm [22], as described in Bartolucci [6]. This algorithm involves alternating between two steps, the E-step and M-step, until convergence. In the E-step, posterior probabilities for each latent class and sample unit are computed based on the individual's response configuration. Subsequently, in the M-step, the expected value of the complete data log-likelihood is maximised using the latent class information from the E-step. To address the possible likelihood function multimodality, we propose repeating the estimation process by randomly selecting initial values for the model parameters. An implementation of the EM algorithm for estimating model (1) is available in the R package MultiLCIRT [23].

However, determining the dimensionality of a latent phenomenon is often challenging due to its unknown nature. Therefore, the approach of identifying the number of dimensions D and determining which variables contribute to measuring each dimension (as indicated by

variables δ_{jd}) typically involves a hierarchical clustering algorithm. This exploratory procedure analyses the latent dimensionality structure of a set of indicators by clustering those that measure the same dimension. It explores all possible clustering combinations for each potential number of dimensions, ideally ranging from 1 to the total number of items J [6].

Initially, the algorithm starts with the most general model, where $D = J$, implying that each item measures a distinct dimension. Subsequently, the algorithm seeks to reduce dimensionality by clustering two items, thereby creating $J - 1$ dimensions. Given $\binom{J}{2}$ possible pairings, multiple models are estimated, and the best model for each dimension count is selected via a likelihood ratio test.

As the algorithm proceeds, it continues to combine items or previously formed clusters, gradually reducing the number of dimensions and evaluating these combinations at each step. This iterative process results in models with progressively fewer dimensions until reaching the unidimensional model, where all items are presumed to measure a single dimension.

These clustering steps can be visualised through a dendrogram, which shows item groupings at each step and the deviance of the model selected at each stage from the initial multidimensional one. Selecting the most appropriate model, and consequently the corresponding number of dimensions, relies on various methods. These could be subjective, based on prior research, or statistical, such as through Information Criteria (IC), which penalise the model's maximum log-likelihood to assess the fit. The model with the lowest IC value is considered optimal, as it achieves a balance between model complexity and fit to the data.

5. Results

In this section we present the main results of our study. Specifically, Section 5.1 illustrates the procedure for selecting the final model, while Section 5.2 discusses the estimates of the selected model, focusing on *i.* the dimensionality structure of corruption risk in healthcare procurement and *ii.* the resulting profiles of healthcare contracting authorities based on the established dimensions. Finally, Section 5.3 describes the profiles on account of individual characteristics of contracting authorities, offering additional insights into their behaviour.

5.1. Model selection

As previously mentioned, given the unknown number of dimensions underlying corruption risk in healthcare public procurement, as well as that of the profiles describing the behaviour of contracting authorities, our investigation entails a full exploratory analysis. This assessment aims to simultaneously search for *i.* the number of dimensions (D) and their composition in terms of red flags that contribute to measuring each one, and *ii.* the number of latent classes (k), corresponding to the profiles of interest accounting for the ascertained dimensions.

The clustering algorithm introduced in Section 4 is therefore run for different values of k , starting from three and increasing it until the model fitting worsens, according to the following penalised log-likelihood indexes: Bayesian Information Criterion (BIC) [24], Akaike Information Criterion (AIC) [25], Consistent AIC (CAIC) [26], AICc [27], and HT-AIC [28]. They can be computed as follows:

$$\text{BIC} = -2\hat{l} + m \log(n),$$

$$\text{AIC} = -2\hat{l} + 2m,$$

$$\text{CAIC} = -2\hat{l} + m(\log(n) + 1),$$

$$\text{AICc} = -2\hat{l} + 2 \frac{m(m-1)}{n-m-1},$$

$$\text{HT-AIC} = -2\hat{l} + 2m + \frac{2(m+1)(m+2)}{n-m-2},$$

where $\hat{l}$ is the maximised model log-likelihood, n is the sample size and m is the number of free parameters.

As stated earlier, for a given number of latent classes k , the algorithm starts (step 0) by fitting the model with one dimension for each indicator (i.e., $D = 14$ dimensions in this case). In the following steps, the procedure seeks the best model with one fewer dimension, until it reaches the final step, where the unidimensional model is considered. Thus, for each value of k , we obtain similar outputs to those presented in Appendix A.1. From these outputs, we retain the best k -specific solution based on the selected information criteria, displayed in Table 4. As we can see, all the indexes except the AIC allow us to stop the clustering algorithm at $k = 9$, as the best solution exhibits IC values greater than those obtained in the previous step (with $k = 8$). Therefore, we select eight latent classes ($k = 8$) as the optimal solution.

Moreover, most of the selection metrics also agree in identifying the final dimensional structure, particularly highlighting that corruption risk in healthcare public procurement can be better represented by four dimensions ($D = 4$).

In light of the aforementioned considerations, we opt for the solution with eight latent classes ($k = 8$) and four dimensions ($D = 4$). The dimensional composition resulting from the clustering algorithm is shown in Fig. 1. The grey dashed line represents the deviance level suggested by the above selection indexes, highlighting four groups of red flags (hence, four dimensions), as follows:

- Group 1: *excl_bids*, *all_bids_excl_but_one*;
- Group 2: *singlebid_num*, *nonopen_econ*, *singlebid_econ*, *nonopen_num*, *eval*, *excl_bids_but_one*;
- Group 3: *MEAT_num*, *MEAT_econ*, *advert*;
- Group 4: *time_dev*, *variants*, *amount_dev*.

To assess the coherent clustering of the 14 red flags within the four dimensions, we can consider discrimination parameters $\hat{\gamma}_j^*$, standardised for cross-dimension comparison purposes. These parameters are obtained using the following formulas [29]:

$$\hat{\gamma}_j^* = \hat{\sigma}_d \hat{\gamma}_j, \quad (2)$$

$$\hat{\sigma}_d = \sqrt{\sum_{c=1}^8 (\hat{\theta}_{cd} - \hat{\mu}_d)^2 \hat{\pi}_c}, \quad (3)$$

$$\hat{\mu}_d = \sum_{c=1}^8 \hat{\theta}_{cd} \hat{\pi}_c, \quad (4)$$

with $j = 1, \dots, 14$ and $d = 1, \dots, 4$. The hat symbol ($\hat{\cdot}$) has the usual meaning of “estimated” value, while $\hat{\sigma}_d$ and $\hat{\mu}_d$ represent estimated standard deviation and mean value of latent trait d .

The standardised discrimination parameters for the initial exploratory solution are reported in Table 5, under the column labelled as “explor.”. This solution identifies two inconsistent red flags with negative discrimination power: *excl_bids_but_one* in Group 2 and *advert* in Group 3. To ensure the robustness of the dimensional clustering, we remove these two inconsistent indicators and proceed with a confirmatory approach using the remaining 12 red flags, while maintaining the same four-dimensional structure established by the exploratory algorithm. The confirmatory approach yields a model with a slightly better (albeit small) absolute fit compared to its exploratory counterpart with 12 red flags (−3015.1 vs. −3015.3; see Appendix A.2 for the complete output about the exploratory procedure with $J = 12$). The confirmatory solution excludes inconsistent or negatively discriminant red flags and provides a clearer interpretation of the identified dimensions, as shown in Table 5 (refer to the last column labelled as “confir.”).

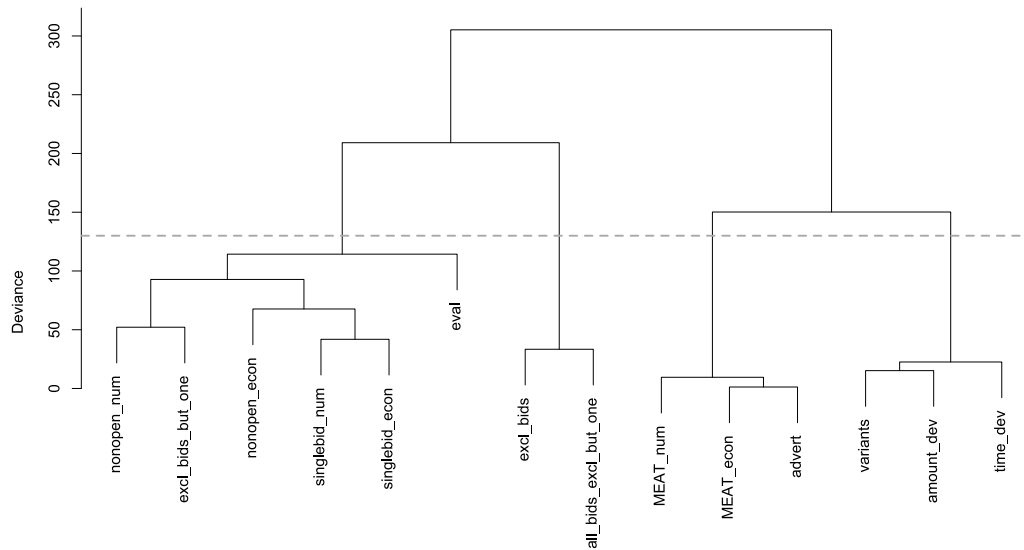
In the following section, we will provide a narrative of each dimension along with a description of the eight latent classes.

5.2. Corruption risk in healthcare public procurement: dimensions of the phenomenon and profiles of contracting authorities

The best model identified by the selection strategy discussed in the previous section is the one that *i.* characterises the corruption

Table 4Model selection (best models resulting from the clustering algorithm with $k = 3, \dots, 9$).

k	BIC	D	AIC	D	CAIC	D	AICc	D	HT-AIC	D
3	7,718.6	2	7,518.4	3	7,778.6	2	7,446.9	3	7,570.8	2
4	7,700.7	4	7,473.5	5	7,767.7	4	7,404.0	5	7,545.4	4
5	7,675.3	3	7,424.3	7	7,744.3	3	7,362.4	5	7,517.7	5
6	7,654.6	4	7,371.6	6	7,731.6	4	7,319.6	6	7,494.3	4
7	7,650.2	5	7,345.1	7	7,735.1	2	7,312.0	5	7,493.0	5
8	7,628.2	4	7,312.1	7	7,713.0	3	7,290.0	4	7,471.0	4
9	7,632.2	3	7,288.4	8	7,717.2	3	7,295.0	4	7,473.6	3

**Fig. 1.** Dendrogram of the clustering algorithm using $k = 8$. Note: the grey dashed line is the selected “cut” level (according to most information criteria), which leads to pinpoint four dimensions.**Table 5**

Dimension composition and discrimination power ($\hat{\gamma}_j^*$) of each red flag: initial exploratory approach with the entire set of red flags – “explor.” ($J = 14$) – and final confirmatory one with a reduced set of indicators – “confir.” ($J = 12$).

Dimension	Red flag	explor. ($J = 14$)	confir. ($J = 12$)
1	<i>excl_bids</i>	4.493	7.034
	<i>all_bids_excl_but_one</i>	4.072	7.374
2	<i>singlebid_num</i>	1.917	1.899
	<i>nonopen_econ</i>	1.150	1.170
	<i>singlebid_econ</i>	1.137	1.174
	<i>nonopen_num</i>	1.087	1.076
	<i>eval</i>	0.288	0.264
	<i>excl_bids_but_one</i>	-0.143	-
3	<i>MEAT_num</i>	2.048	1.996
	<i>MEAT_econ</i>	1.776	1.826
	<i>advert</i>	-0.502	-
4	<i>time_dev</i>	1.187	1.227
	<i>variants</i>	1.163	1.140
	<i>amount_dev</i>	0.421	0.418

risk in healthcare public procurement into four dimensions, hence four different types of risk ($D = 4$) and, at the same time, *ii.* identifies eight different clusters of healthcare contracting authorities ($k = 8$), to be characterised based on the four dimensions.

As far as the dimensionality assessment is concerned, we can examine the final composition of each dimension, as reported in [Table 5](#) under the column labelled as “confir.”. The first dimension encompasses red flags referring to risks related to: *i.* the circumstance where a contracting authority excludes all but one offer during bid evaluation (red flag: *all_bids_excl_but_one*); *ii.* the proportion of

excluded bids in procedures with all bids excluded except one (red flag: *excl_bids_but_one*).

The second dimension groups red flags referring to risks associated with: *i.* instances where a procurement procedure attracts a single bid, both in relative terms and in economic value (red flags: *singlebid_num* and *singlebid_econ*); *ii.* the type of procurement procedure, including non-open procedures such as direct assignments or negotiated procedures, both in relative terms and economic value (red flags: *nonopen_num* and *nonopen_econ*); *iii.* the time needed by the contracting authority to complete the bid evaluation process, measured as the time elapsed between bid submission and award notice (red flag: *eval*). Standardised discrimination parameters indicate that the red flags contributing most to characterising this dimension are those related to the single bidding, with non-open procurement procedures also playing a significant role. The red flag with the lowest contribution to this dimension is *eval*, as indicated by its discrimination value.

The third dimension comprises red flags referring to risks associated with the contract award criterion or the weight of procedures awarded with discretionary criteria (i.e., the most economically advantageous tender), both in relative terms and in economic value (red flags: *MEAT_num* and *MEAT_econ*).

The fourth dimension groups red flags highlighting risks related to: *i.* the occurrence of variations during contract implementation affecting its duration, computed as the ratio between real and envisaged duration, averaged over all procedures (red flag: *time_dev*); *ii.* the proportion of procedures with at least one variant out of all awarded procedures (red flag: *variants*); *iii.* the occurrence of variations during a contract implementation affecting its awarded value, that is, the (average) extent of the variation between awarded and cleared contract value (red flag: *amount_dev*).

Overall, the first two identified risk dimensions underscore the absence of competitiveness as the primary mechanism undermining the

fairness and integrity of public procurement and contributing to corruption risks. The absence of competition has the potential to concentrate wealth and influence among a small group of dominant players and to lead to increased prices, diminished consumer well-being, reduced product variety, and restricted innovation.

In the first dimension, access restriction to procurement procedures occurs through the exclusion of potential competitors, facilitated by the key role played by contracting authorities. Thus, lack of competition primarily arises from the discretionary power exerted by contracting authority during the contract awarding phase. Indeed, excess of discretion may allow them to shape procurement processes in a manner that favours certain companies while excluding others, potentially fostering an environment conducive to corruption and anti-competitive behaviour.

Differently, in the second dimension, collusive agreements among companies participating in the tender take on the main role, alongside the role of contracting authorities. Here, the success of a particular bidder in obtaining a contract is most likely the result of bid rigging (i.e., companies agreeing in advance who will submit the winning bid or rotating winning bids among them) and/or bid suppression (i.e., companies agreeing in advance not to bid on certain contracts, allowing a specific company to win uncontested).

The third dimension identifies risks mostly related to the contract award criterion and emphasises the excess of discretionary power exerted by contracting authorities when assessing offers as the main driver of possible corruption risks. Indeed, an excess of discretionary power can undermine the principles of transparency, weaken disclosure and accountability mechanisms (i.e., public consultations, complaint mechanisms, audit processes, etc.), and lead to conflict of interest, favouritism, and the misuse of resources.

The fourth dimension identifies risks related to the occurrence of variations during contract implementation affecting its actual duration and awarded economic value. Inefficiency in public procurement management is the key driver in this dimension, as delays in project implementation and cost overruns may create incentives for officials to engage in corrupt practices to expedite processes or cover up inefficiencies.

Based on the considerations above, the four dimensions of corruption risk are therefore synthetically categorised as:

- Dimension 1: Corruption risks associated with lack of competition induced primarily by contracting authorities over the tendering phase;
- Dimension 2: Corruption risks associated with lack of competition induced primarily by collusive agreements between companies and contracting authorities;
- Dimension 3: Corruption risks associated with excess of discretionary power exerted by contracting authorities over pre-tendering;
- Dimension 4: Corruption risks associated with inefficiencies in public procurement project implementation.

Grounded on the above four dimensions of corruption risks, we can proceed with qualifying the eight groups (latent classes, LC) of contracting authorities in the health sector and including, as mentioned earlier, local health authorities, hospital administrations, and university hospital administrations. To better describe the eight profiles, we employ estimated standardised latent trait levels, $\hat{\theta}_{cd}^*$, together with the prior probability of each latent class, $\hat{\pi}_c$, with $c = 1, \dots, 8$ and $d = 1, \dots, 4$. Standardised latent traits are computed as $\hat{\theta}_{cd}^* = (\hat{\theta}_{cd} - \hat{\mu}_d) / \hat{\sigma}_d$, where $\hat{\mu}_d$ and $\hat{\sigma}_d$ are computed as in (3) and (4), respectively. The standardised latent trait levels are shown in Table 6 and represented in Fig. 2.

As it can be appreciated from the picture, the first latent class (LC1) consists of contracting authorities for which the risk of corruption is especially high on the second dimension, related to lack of competition induced primarily by collusive agreements between companies

and contracting authorities. The second latent class (LC2) clusters contracting authorities for which the risk of corruption is medium-high on both the second and third dimension (associated to excess of discretionary power exerted by contracting authorities over pre-tendering). Latent class 3 (LC3) and latent class 4 (LC4) exhibit similar risk profiles across the first three dimensions but differ in dimension 4, highlighting corruption risks associated with inefficiencies in project implementation. Specifically, LC3 comprises contracting authorities for which the risk of corruption is especially high on this fourth dimension, whereas LC4 exhibits lower risks associated with inefficiencies. Latent class 5 (LC5) encompasses contracting authorities for which the risk of corruption is medium-high on the third dimension, highlighting risks associated to excess of discretionary power exerted by contracting authorities over pre-tendering. Latent class 6 (LC6) is characterised by a medium-high level of corruption over dimension 1, related to a lack of competition induced primarily by contracting authorities over the tendering phase, and dimension 4, related to corruption risks associated with inefficiencies. Latent class 7 (LC7) exhibits a high level of corruption especially over dimension 3, related to risks stemming from the excessive discretionary power wielded by contracting authorities during the pre-tendering phase. Finally, latent class 8 (LC8) shows the highest risk on dimension 1, related to lack of competition primarily induced by contracting authorities during the tendering process, and medium levels of corruption risks over dimensions 2, 3, and 4.

In terms of their prevalence, examining prior probabilities $\hat{\pi}_c$ in Table 6, we can observe that LC5 and LC2 are the most numerous, representing approximately 40% of healthcare contracting authorities, while LC6 and LC8 are numerically the smallest.

5.3. Features of contracting authority profiles

This section aims to assess the potential relationship between the eight corruption risk profiles and individual characteristics of healthcare contracting authorities encompassed within them. To achieve this, we examine the (posterior) latent class allocation of each authority and explore whether there is any association between these allocations and the individual characteristics of the authorities. These characteristics include:

- territorial distribution over the 20 Italian regions (NUTS-2 level);
- typology: *i.* regional health agencies, *ii.* local health authorities, and *iii.* public hospital trusts, university hospital trusts, polyclinics, and institutes for hospitalisation and scientific treatment;
- size, computed as the average number of employees of each contracting authority (over the time period considered for this work) using data from “Conto Annuale”, an open data platform handled by the Italian Ministry of Economy and Finance.²

As for the territorial distribution, the association with the latent class allocation – assessed using the Fisher’s exact test for count data – is significant (with p -value < 0.001). In this regard, readers can refer to Fig. 3, which presents eight maps of Italy, each corresponding to a different risk profile. The maps display the territorial distribution of contracting authorities across Italian regions, with colours representing the frequency of authorities located in each one.

Latent class 1 (LC1), which indicates a risk profile characterised by a lack of competition primarily due to collusive agreements between companies and contracting authorities, shows a relatively uniform territorial distribution. However, there is a slight concentration in the northern regions, such as Lombardia and Friuli-Venezia Giulia. Latent class 3 – where the risk of corruption due to inefficiencies is the highest – is mostly present in Lombardia and to a smaller extent in Veneto, Emilia-Romagna and Tuscany. As regards LC7 – associated with a

² <https://contoannuale.rgs.mef.gov.it>

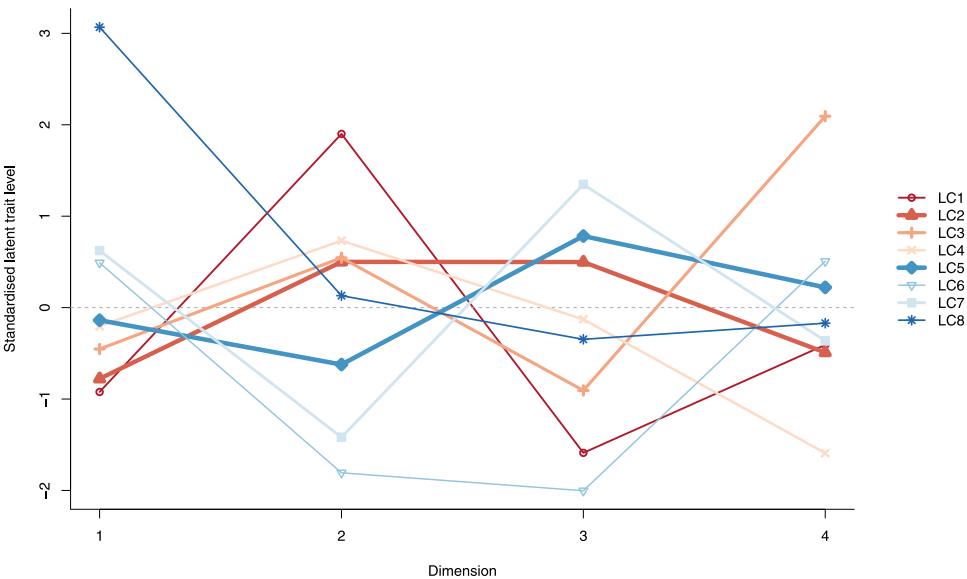


Fig. 2. Line plots of the eight profiles of healthcare contracting authorities (LC1, ..., LC8) by dimensions of corruption risk in healthcare procurement (line width reflects the weight of each profile in terms of prior probability $\hat{\pi}_c$).

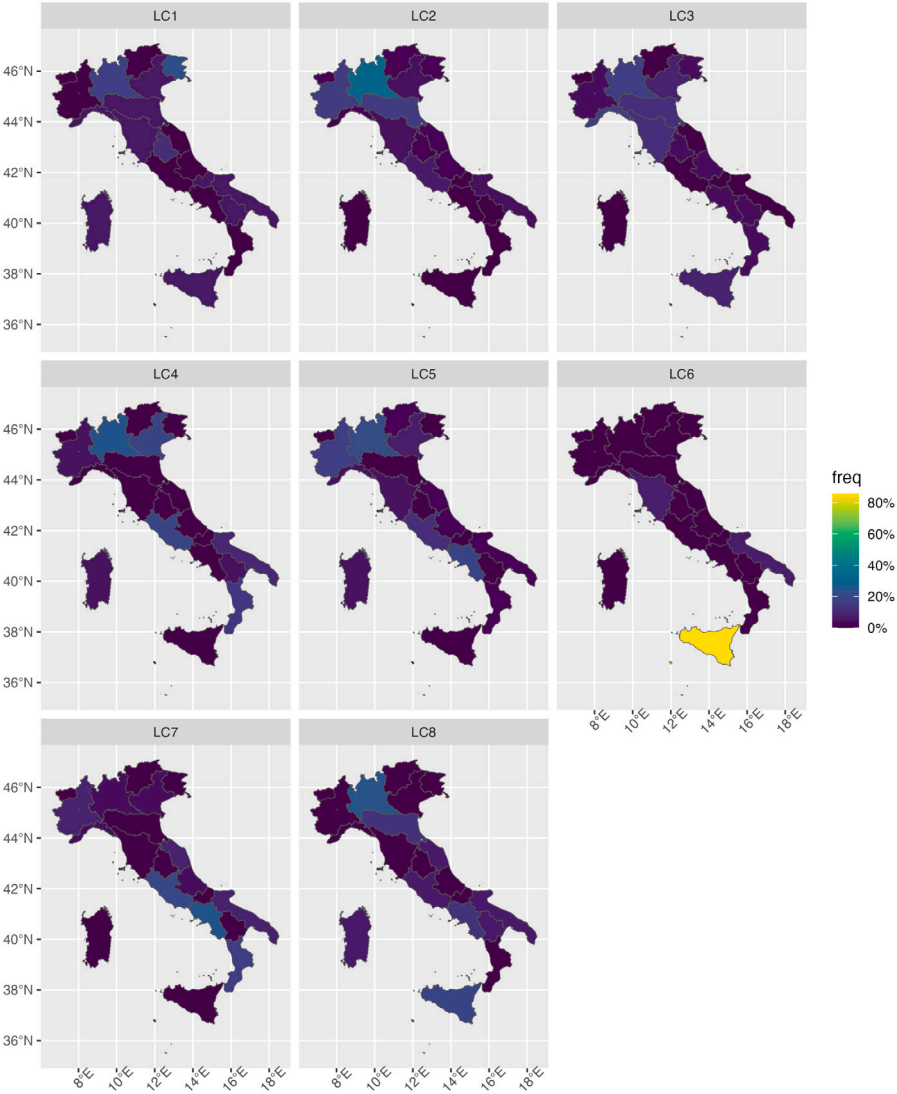


Fig. 3. Territorial distribution of the eight risk profiles in healthcare procurement.

Table 6
Standardised latent trait and prior probability estimates.

Dimension	LC1	LC2	LC3	LC4	LC5	LC6	LC7	LC8
1	−0.922	−0.778	−0.453	−0.203	−0.138	0.491	0.624	3.068
2	1.900	0.501	0.545	0.732	−0.622	−1.807	−1.418	0.131
3	−1.589	0.499	−0.906	−0.128	0.783	−2.003	1.348	−0.347
4	−0.407	−0.491	2.094	−1.592	0.221	0.507	−0.363	−0.170
$\hat{\pi}_c$	0.083	0.195	0.139	0.112	0.202	0.065	0.128	0.075

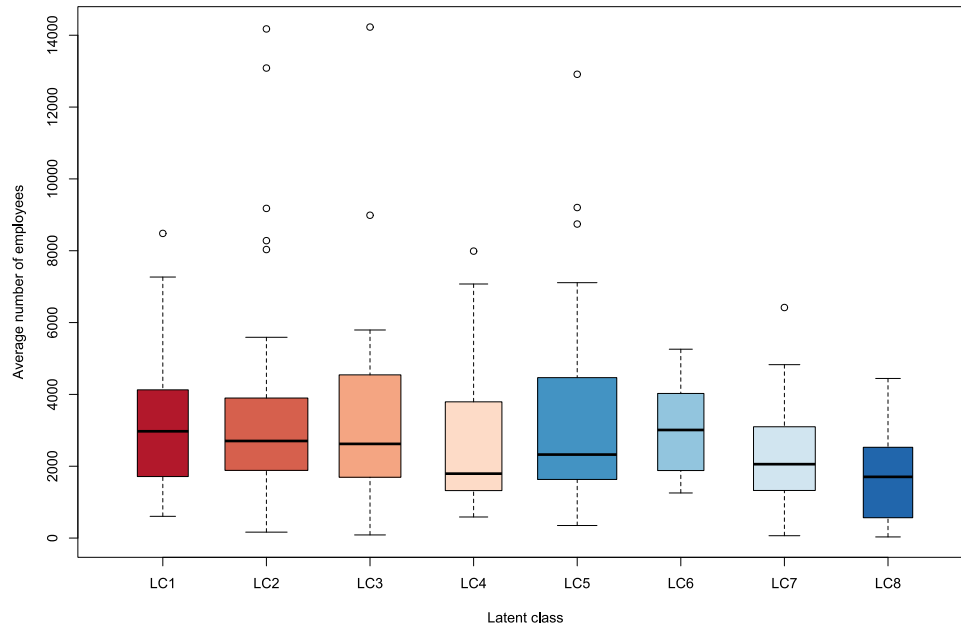


Fig. 4. Distribution of the number of employees in each profile of healthcare contracting authorities (box basis width reflects the weight of each class in terms of prior probability $\hat{\pi}_c$).

high degree of discretionary power exercised by contracting authorities – there is a notable presence in the central-southern regions of the country, especially in Campania and Lazio, which account for approximately 25% and 20% of the contracting authorities, respectively. LC8, characterised by the highest risk due to a lack of competition caused by contracting authorities over the tendering stage, stands out particularly in Lombardia and Sicilia, where it represents approximately 45% of the health authorities clustered in this group. Finally, LC6 is almost exclusively found in Sicilia. This last result implies that in the Sicilian region the risk of corruption due to lack of competition and to inefficiencies is one among the highest.

As regards the healthcare authority typology, the association with the latent class classification results to be not significant (Fisher's exact test p -value = 0.882).

Finally, the analysis of the association between latent class assignment and contracting authority size is ran to assess whether the distribution of authority sizes differs across latent classes. Upon qualitative examination of the boxplots presented in Fig. 4, shifts in the distributions appear to be minor across the eight profiles. Besides, the two-sample Kolmogorov–Smirnov test is employed to further assess whether the distributions of authority sizes are significantly different between each pair of latent classes (28 pairs overall). Only two pairs yield significant (or near-significant) results: LC8 vs. LC6 (p -value = 0.021) and LC8 vs. LC5 (p -value = 0.054). This latest result implies that latent class 8 – where the risk of corruption due to lack of competition primarily induced by contracting authorities during the

tendering process is the highest – is composed of contracting authorities whose size is significantly smaller than that of authorities grouped in latent class 6 and 5.

6. Discussion of main results

In this paper, we cluster contracting authorities responsible for procuring goods and services for healthcare facilities based on inter-related sub-dimensions of corruption risk. Using a pool of 14 red flag indicators of corruption risk in public procurement, we analyse big data from the Italian National Database of Public Contracts (BDNCP) through an extended Item Response Theory (IRT) model.

In line with both our expectations and previous studies [17], the results demonstrate that corruption risk cannot be viewed as a one-dimensional phenomenon. Instead, it manifests as a multidimensional occurrence, characterised by various behaviours, facts, and circumstances that differ in nature, kind, and intensity. Within the selected four-dimensional solution, the identified dimensions of corruption risks include: *i.* corruption risks stemming from a lack of competition primarily influenced by contracting authorities during the tendering phase; *ii.* corruption risks arising from a lack of competition primarily driven by collusive agreements among companies and contracting authorities; *iii.* corruption risks linked to an excess of discretionary power exercised by contracting authorities during the pre-tendering phase; *iv.* corruption risks associated with inefficiencies in public procurement project implementation.

The literature widely recognises lack of competition, excessive discretionary power, and inefficiencies as key conditions that exacerbate corruption in public procurement. Indeed, corruption in public procurement occurs when public officials deviate from legally mandated principles of open and fair competition during the design and execution of tenders, leading to the recurrent awarding of contracts to affiliated or favoured companies [14,30]. The primary objective of corruption in public procurement is to manipulate contract awards in favour of preferred bidders while avoiding detection. This goal can be achieved through various means, including circumventing competition via direct awards [31,32]. Further, situations where public officials wield excessive or unchecked authority and freedom to make discretionary choices throughout the procurement process can significantly intensify the likelihood of corruption and favouritism. Such situations not only compromise transparency and fairness but also erode competition within public procurement procedures [33,34]. Inefficiencies can heighten the risk of corruption in public procurement as well. Procurement processes that experience significant delays in decision-making, contract award, and delivery of goods and services, as well as those resulting in inflated or substandard quality, can obscure corrupt agreements. Moreover, such inefficiencies may incentivise officials to engage in further malpractices to expedite processes or conceal inefficiencies [33,35].

On the ground of the four dimensions of corruption risks outlined above, we qualify eight latent classes of contracting authorities within the health sector. The findings reveal that each identified cluster of contracting authorities have a unique and clearly defined risk profile, distinct from that of the other clusters. None of the eight groups of contracting authorities consistently display a high level of corruption risk across all four dimensions. Instead, their behaviour vary across different risky circumstances. Overall, the cluster of contracting authorities exhibiting the lowest level of risk across the four dimensions is latent class 6 (LC6). Corruption risks associated with lack of competition induced primarily by contracting authorities over the tendering phase are notably high in latent class 8 (LC8), while risks stemming from a lack of competition driven by collusive agreements between companies and contracting authorities are particularly pronounced in latent class 1 (LC1). Latent class 7 (LC7) comprises contracting authorities at risk of being corrupt mostly on account of an excess of discretionary power during the pre-tendering phase. On the other hand, corruption risks associated with inefficiencies in public procurement project implementation are notably high in latent class 3 (LC3).

The analysis conducted to assess the association between latent class allocation of contracting authorities and their geographical location provides detailed insights into the distribution of corruption risk profiles across Italian regions (NUTS-2 level). Although the overall distribution appears relatively uniform, certain latent classes show a marked prevalence in specific regions, suggesting the existence of distinct regional dynamics in the procurement of healthcare medical equipment. Specifically, in Lombardia and, to a similar extent in Sicilia, the noticeable presence of certain risk profiles suggests a complex and potentially problematic procurement landscape. In particular, the relevant presence of latent class 8 (LC8) – marked by the highest risk related to the lack of competition induced by contracting authorities, alongside medium levels of risk associated with collusive agreements and excessive discretionary power – implies that these Italian regions might be facing multifaceted procurement challenges. What is more, the variety of prevalent risk profiles observed especially in Lombardia may imply the presence of multiple risk factors and the need for comprehensive anti-corruption measures in this region. Finally, the high occurrence of LC6 in Sicilia reflects a regional procurement environment where not only competition is restricted, but also inefficiencies in the management of the procurement processes are high, raising concerns about the integrity and fairness of tendering processes in this region. These observations imply that while some regions may require targeted interventions to address specific aspects of corruption risk,

others – such as Lombardia – might benefit from a broader range of anti-corruption strategies.

The analysis of the relationship between the eight corruption risk profiles and type and size of healthcare contracting authorities show generally negligible results, with the only exception of the association between two pairs of latent classes (LC8 vs. LC6 and LC8 vs. LC5) and their relative size, suggesting that smaller authorities may face higher corruption risks related to insufficient competition during the tendering process.

Overall, the analysis reveals the complexities of corruption risks in healthcare procurement and provides insights into distinct risk profiles among Italian contracting authorities. Indeed, the study findings demonstrate important geographic variations in corruption risk profiles among Italian regions and underscore the importance of customised anti-corruption strategies to address region-specific challenges and risk factors effectively. In this regard, future developments of the current work are related to the possibility of deepening the characterisation of the ascertained profiles by *i.* incorporating covariates of healthcare contracting authorities directly in an extended version of the current model, and *ii.* considering the longitudinal component of this phenomenon, hence relying on hidden Markov models [36,37] for analysing the transition over time across risk profiles.

Despite its potential, we caution the reader to be aware of the potential issue of false positives in the adopted approach. This situation may occur, for instance, when a procurement procedure experiences genuine unexpected circumstances necessitating a pause in the bidding process, a reassessment of project specifications, and the issuance of variants modifying the contract initial economic value. Additionally, in the dimensional solution presented, the inefficiency dimension cannot disentangle between active waste, characterised by deliberate and intentional actions resulting in the inefficient use of public resources within the procurement process, and passive waste, which arises from negligence, incompetence, or oversight rather than deliberate misconduct [38,39]. This limitation may as well lead to an overestimation of the risk of corruption.

Finally, a particular mention is devoted to missingness. Absence of information on contract features warrants attention, as it could indicate a deliberate effort by contracting authorities to withhold tender details. However, in this paper we have chosen to focus solely on contracts where all the necessary features for computing a specific red flag are present in the data. By doing so, our latent variable effectively addresses corruption risk by focussing on the declared tender characteristics, excluding procedures with undeclared or missing information. Consequently, given the difficulty in distinguishing voluntary from involuntary omissions, we have opted to treat this missing mechanism as ignorable.

We acknowledge that our hypothesis oversimplifies real-world complexities. Addressing non-ignorable missing data would require the model to incorporate an additional latent variable, possibly correlated with the primary variable accounting for corruption risk. Bacci and Bartolucci [20] propose a similar approach for binary data in the educational context, which could serve as a foundation. Extending this model methodologically would be necessary to accommodate our non-binary or mixed-type items. Moreover, this extension should also account for the hierarchical structure of the data, managing two nested target units (contracts managed by contracting authorities).

CRedit authorship contribution statement

Simone Del Sarto: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Michela Gnaldi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Niccolò Salvini:** Writing – original draft, Visualization, Methodology.

Declaration of competing interest

No potential competing interest was reported by the authors.

Data availability

The data that has been used is confidential.

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Appendix

A.1. Exploratory clustering algorithm with $J = 14$

In this section we include tables regarding the initial exploratory analyses on the dimensionality assessment, obtained running the clustering algorithm on the entire set of red flags ($J = 14$) with different values of k (up to 9). The best solutions according to each information criterion are reported in boldface.

$k = 3$								
Step	$\hat{I}$	m	D	BIC	AIC	CAIC	AICc	HT-AIC
0	-3,696.2	72	14	7,776.4	7,536.5	7,848.4	7,468.8	7,617.7
1	-3,696.2	71	13	7,771.1	7,534.5	7,842.1	7,466.1	7,612.9
2	-3,696.2	70	12	7,765.8	7,532.5	7,835.8	7,463.5	7,608.2
3	-3,696.2	69	11	7,760.5	7,530.5	7,829.5	7,461.0	7,603.6
4	-3,696.3	68	10	7,755.1	7,528.5	7,823.1	7,458.5	7,599.0
5	-3,696.3	67	9	7,749.9	7,526.6	7,816.9	7,456.2	7,594.6
6	-3,696.4	66	8	7,744.7	7,524.7	7,810.7	7,454.0	7,590.3
7	-3,696.4	65	7	7,739.5	7,522.9	7,804.5	7,451.9	7,586.0
8	-3,696.7	64	6	7,734.6	7,521.3	7,798.6	7,450.1	7,582.2
9	-3,696.9	63	5	7,729.8	7,519.9	7,792.8	7,448.5	7,578.5
10	-3,697.2	62	4	7,725.1	7,518.5	7,787.1	7,447.0	7,574.9
11	-3,698.2	61	3	7,721.7	7,518.4	7,782.7	7,446.9	7,572.7
12	-3,699.3	60	2	7,718.6	7,518.7	7,778.6	7,447.1	7,570.8
13	-3,706.3	59	1	7,727.2	7,530.6	7,786.2	7,459.2	7,580.8

$k = 4$								
Step	$\hat{I}$	m	D	BIC	AIC	CAIC	AICc	HT-AIC
0	-3,664.1	87	14	7,792.2	7,502.3	7,879.2	7,454.0	7,635.0
1	-3,664.3	85	13	7,781.8	7,498.5	7,866.8	7,446.6	7,623.2
2	-3,664.6	83	12	7,771.8	7,495.2	7,854.8	7,439.9	7,612.3
3	-3,664.7	81	11	7,761.4	7,491.5	7,842.4	7,433.2	7,601.3
4	-3,665.0	79	10	7,751.3	7,488.0	7,830.3	7,427.1	7,590.9
5	-3,665.3	77	9	7,741.3	7,484.7	7,818.3	7,421.4	7,581.0
6	-3,665.8	75	8	7,731.5	7,481.5	7,806.5	7,416.3	7,571.6
7	-3,666.2	73	7	7,721.8	7,478.5	7,794.8	7,411.5	7,562.6
8	-3,666.8	71	6	7,712.3	7,475.7	7,783.3	7,407.3	7,554.1
9	-3,667.7	69	5	7,703.4	7,473.5	7,772.4	7,404.0	7,546.5
10	-3,671.7	67	4	7,700.7	7,477.4	7,767.7	7,407.1	7,545.4
11	-3,679.4	65	3	7,705.4	7,488.8	7,770.4	7,417.8	7,552.0
12	-3,688.8	63	2	7,713.6	7,503.7	7,776.6	7,432.3	7,562.3
13	-3,925.6	61	1	8,176.5	7,973.2	8,237.5	7,901.7	8,027.5

$k = 5$								
Step	$\hat{I}$	m	D	BIC	AIC	CAIC	AICc	HT-AIC
0	-3,622.9	102	14	7,789.8	7,449.9	7,891.8	7,444.0	7,657.9
1	-3,623.0	99	13	7,774.0	7,444.1	7,873.0	7,427.4	7,634.6
2	-3,623.2	96	12	7,758.3	7,438.4	7,854.3	7,412.2	7,612.8
3	-3,623.8	93	11	7,743.5	7,433.6	7,836.5	7,399.0	7,593.1
4	-3,624.8	90	10	7,729.5	7,429.6	7,819.5	7,387.7	7,575.2
5	-3,626.2	87	9	7,716.4	7,426.4	7,803.4	7,378.2	7,559.2
6	-3,628.2	84	8	7,704.4	7,424.4	7,788.4	7,370.7	7,545.3
7	-3,631.2	81	7	7,694.3	7,424.3	7,775.3	7,366.0	7,534.1
8	-3,634.8	78	6	7,685.5	7,425.6	7,763.5	7,363.4	7,525.1
9	-3,638.8	75	5	7,677.6	7,427.7	7,752.6	7,362.4	7,517.7
10	-3,648.5	72	4	7,681.0	7,441.0	7,753.0	7,373.3	7,522.2
11	-3,653.7	69	3	7,675.3	7,445.4	7,744.3	7,375.8	7,518.4
12	-3,665.3	66	2	7,682.5	7,462.5	7,748.5	7,391.8	7,528.1
13	-3,925.6	63	1	8,187.2	7,977.2	8,250.2	7,905.9	8,035.8

$k = 6$								
Step	$\hat{I}$	m	D	BIC	AIC	CAIC	AICc	HT-AIC
0	-3,585.3	117	14	7,794.5	7,404.5	7,911.5	7,475.5	7,723.7
1	-3,585.5	113	13	7,773.6	7,397.0	7,886.6	7,443.2	7,682.0
2	-3,586.9	109	12	7,755.0	7,391.7	7,864.0	7,416.4	7,646.1
3	-3,589.2	105	11	7,738.3	7,388.4	7,843.3	7,394.6	7,615.2
4	-3,592.1	101	10	7,722.8	7,386.2	7,823.8	7,376.6	7,588.3
5	-3,592.4	97	9	7,702.1	7,378.8	7,799.1	7,355.7	7,558.5
6	-3,594.0	93	8	7,683.8	7,373.9	7,776.8	7,339.3	7,533.4
7	-3,597.2	89	7	7,669.1	7,372.5	7,758.1	7,328.4	7,513.7
8	-3,600.8	85	6	7,654.9	7,371.6	7,739.9	7,319.6	7,496.3
9	-3,617.3	81	5	7,666.5	7,396.5	7,747.5	7,338.2	7,506.3
10	-3,622.0	77	4	7,654.6	7,398.0	7,731.6	7,334.7	7,494.3
11	-3,635.2	73	3	7,659.8	7,416.5	7,732.8	7,349.5	7,500.6
12	-3,650.4	69	2	7,668.8	7,438.8	7,737.8	7,369.3	7,511.9
13	-3,925.6	65	1	8,197.9	7,981.2	8,262.9	7,910.3	8,044.4

$k = 7$								
Step	$\hat{I}$	m	D	BIC	AIC	CAIC	AICc	HT-AIC
0	-3,555.4	132	14	7,814.8	7,374.9	7,946.8	7,578.2	7,863.1
1	-3,556.1	127	13	7,789.4	7,366.1	7,916.4	7,517.2	7,789.5
2	-3,558.4	122	12	7,767.4	7,360.8	7,889.4	7,468.3	7,728.3
3	-3,559.9	117	11	7,743.8	7,353.8	7,860.8	7,424.8	7,673.0
4	-3,562.3	112	10	7,721.9	7,348.6	7,833.9	7,389.1	7,625.7
5	-3,566.4	107	9	7,703.5	7,346.9	7,810.5	7,362.0	7,587.1
6	-3,571.6	102	8	7,687.1	7,347.2	7,789.1	7,341.3	7,555.2
7	-3,575.6	97	7	7,668.4	7,345.1	7,765.4	7,322.0	7,524.8
8	-3,584.8	92	6	7,660.3	7,353.7	7,752.3	7,316.6	7,508.4
9	-3,593.1	87	5	7,650.2	7,360.2	7,737.2	7,312.0	7,493.0
10	-3,614.2	82	4	7,665.7	7,392.4	7,747.7	7,335.5	7,505.7
11	-3,631.5	77	3	7,673.6	7,417.0	7,750.6	7,353.7	7,513.3
12	-3,639.6	72	2	7,663.1	7,423.2	7,735.1	7,355.5	7,504.4
13	-3,925.6	67	1	8,208.5	7,985.2	8,275.5	7,914.9	8,053.2

$k = 8$								
Step	$\hat{I}$	m	D	BIC	AIC	CAIC	AICc	HT-AIC
0	-3,525.0	147	14	7,833.8	7,343.9	7,980.8	7,777.4	8,104.3
1	-3,525.6	141	13	7,803.1	7,333.2	7,944.1	7,658.6	7,967.7
2	-3,529.7	135	12	7,779.4	7,329.5	7,914.4	7,569.0	7,861.8
3	-3,532.6	129	11	7,753.0	7,323.1	7,882.0	7,494.0	7,771.3
4	-3,536.2	123	10	7,728.4	7,318.5	7,851.4	7,434.1	7,696.5
5	-3,541.6	117	9	7,707.2	7,317.3	7,824.2	7,388.3	7,636.4
6	-3,545.9	111	8	7,683.7	7,313.8	7,794.7	7,348.8	7,583.0
7	-3,551.0	105	7	7,662.0	7,312.1	7,767.0	7,318.3	7,538.9
8	-3,558.8	99	6	7,645.5	7,315.6	7,744.5	7,298.9	7,506.1
9	-3,571.4	93	5	7,638.7	7,328.7	7,731.7	7,294.1	7,488.2
10	-3,582.1	87	4	7,628.2	7,338.2	7,715.2	7,290.0	7,471.0
11	-3,600.0	81	3	7,632.0	7,362.1	7,713.0	7,303.8	7,471.9
12	-3,629.5	75	2	7,659.0	7,409.1	7,734.0	7,343.8	7,499.1
13	-3,677.6	69	1	7,723.1	7,493.1	7,792.1	7,423.6	7,566.2

$k = 9$								
Step	$\hat{I}$	m	D	BIC	AIC	CAIC	AICc	HT-AIC
0	-3,499.4	162	14	7,862.7	7,322.8	8,024.7	8,184.3	8,566.1
1	-3,499.8	155	13	7,826.2	7,309.7	7,981.2	7,935.7	8,289.3
2	-3,501.9	148	12	7,793.1	7,299.9	7,941.1	7,754.1	8,084.1
3	-3,509.7	141	11	7,771.3	7,301.4	7,912.3	7,626.8	7,935.9
4	-3,511.9	134	10	7,738.4	7,291.9	7,872.4	7,518.9	7,809.0
5	-3,520.0	127	9	7,717.3	7,294.1	7,844.3	7,445.2	7,717.4
6	-3,524.2	120	8	7,688.3	7,288.4	7,808.3	7,380.5	7,635.7
7	-3,535.6	113	7	7,673.8	7,297.2	7,786.8	7,343.4	7,582.2
8	-3,543.7	106	6	7,652.6	7,299.3	7,758.6	7,309.9	7,532.8
9	-3,557.9	99	5	7,643.7	7,313.8	7,742.7	7,297.1	7,504.4
10	-3,574.1	92	4	7,638.8	7,332.2	7,730.8	7,295.0	7,486.9
11	-3,589.5	85	3	7,632.2	7,348.9	7,717.2	7,296.9	7,473.6
12	-3,624.1	78	2	7,664.2	7,404.2	7,742.2	7,342.0	7,503.7
13	-3,675.9	71	1	7,730.5	7,493.9	7,801.5	7,425.5	7,572.3

A.2. Exploratory clustering algorithm with $J = 12$

In this section we show the results of the exploratory clustering algorithm, run during the robustness check of the final dimensional solution, using $k = 8$ and a reduced set of red flags ($J = 12$). The best solutions according to each information criterion are reported in boldface.

Step	$\hat{I}$	m	D	BIC	AIC	CAIC	AICc	HT-AIC
0	-2,973.0	127	12	6,623.2	6,199.9	6,750.2	6,351.0	6,623.3
1	-2,974.9	121	11	6,595.0	6,191.7	6,716.0	6,291.4	6,549.0
2	-2,976.1	115	10	6,565.6	6,182.3	6,680.6	6,240.4	6,483.9
3	-2,981.1	109	9	6,543.5	6,180.3	6,652.5	6,205.0	6,434.6
4	-2,987.0	103	8	6,523.2	6,180.0	6,626.2	6,178.0	6,394.1
5	-2,988.6	97	7	6,494.5	6,171.3	6,591.5	6,148.1	6,350.9
6	-2,997.0	91	6	6,479.2	6,175.9	6,570.2	6,136.4	6,326.0
7	-3,010.5	85	5	6,474.2	6,190.9	6,559.2	6,138.9	6,315.6
8	-3,015.3	79	4	6,451.8	6,188.5	6,530.8	6,127.5	6,291.4
9	-3,036.8	73	3	6,462.8	6,219.5	6,535.8	6,152.6	6,303.6
10	-3,072.1	67	2	6,501.4	6,278.1	6,568.4	6,207.7	6,346.1
11	-3,351.9	61	1	7,029.2	6,825.9	7,090.2	6,754.4	6,880.1

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